

5.01 The commutative and associative laws

(booklet p. 3)

WE DO examples

- Ex 1: 28
- Ex 2: 118
- Ex 3: 170
- Ex 4: 3 100
- Ex 5: 390
- Ex 6: 2 800

Set A

- 28
- 118
- 75
- 58
- 117
- 50

Set B

- 170
- 3 100
- 900
- 1 300
- 700
- 400

Set C

- 390
- 2 800
- 600
- 2 400
- 2 400
- 300

Set D

- No: $10 - 4 = 6$ but $4 - 10 = -6$
- No: $(24 \div 6) \div 2 = 2$ but $24 \div (6 \div 2) = 8$
- 17 000
- 210 (10 pairs of 21)

Extension

- 90 000
- $38 + 62 = 100$ and $47 + 53 = 100$

Exit ticket

- Level 1: 47
- Level 2: 230
- Level 3: 1 300

5.02 The distributive law (booklet p. 7)

WE DO examples

- Ex 1: 198
- Ex 2: 136
- Ex 3: 297
- Ex 4: 156
- Ex 5: 288
- Ex 6: 3 465

Set A

- 84
- 92
- 217
- 95
- 126
- 116

Set B

- 198
- 136
- 297
- 156
- 105
- 117

Set C

- 288
- 3 465
- 2 323
- 636
- 1 592
- 1 365

Set D

- $7 \times 100 = 700$
- $15 \times 10 = 150$
- e.g. $18 \times 25 = 10 \times 25 + 8 \times 25 = 450$; or $20 \times 25 - 2 \times 25 = 450$
- $97 = 100 - 3$, so multiply each part by 6

Extension

- 64 064
- 3 700

Exit ticket

- Level 1: 84
- Level 2: 114
- Level 3: 4 590

5.03 Variables and algebraic notation

(booklet p. 11)

WE DO examples

- Ex 1: $a - (-a) = 2 \times a$
- Ex 2: $4x$
- Ex 3: $10a$
- Ex 4: $24abc$
- Ex 5: $4 \times m \times n \times k$
- Ex 6: $9 \times c \times c$, $6 \times a \div b$

Set A

- $4x$
- $\frac{u}{5}$
- y
- $10a$
- $7m$
- pq

Set B

- $24abc$
- $20mw^2$
- k^3
- $3y^2$
- $2a + 3b$
- $2d$

Set C

1. $4 \times m \times n \times k$
2. $6 \times a \div b$
3. $9 \times c \times c$
4. $3 \times x \times y \times y$
5. $a \times a \times a \times b$
6. $5 \times m \div (2 \times n)$

Set D

1. $a + a = 2 \times a$
2. $a \times 0 = 0$
3. $3n + 1$ sticks
4. No: $3a^2 = 3 \times a \times a$ but $(3a)^2 = 9 \times a \times a$

Extension

1. $n \times n - (n - 1)(n + 1) = 1$
2. check

Exit ticket

1. Level 1: $3y$
2. Level 2: $12pq$
3. Level 3: $5 \times a \times a \times b$

5.04 Expressions from words (booklet p. 15)

WE DO examples

1. Ex 1: $3N$
2. Ex 2: $N - 5$
3. Ex 3: $N + 1$
4. Ex 4: $\frac{N}{3}$
5. Ex 5: $2H - k$
6. Ex 6: $2(x + 4)$

Set A

1. $N + 7$
2. $3N$
3. $N - 5$
4. $\frac{N}{4}$
5. $N + 1$
6. $\frac{N}{3}$

Set B

1. $M + 3$
2. $B + 5$
3. $2H - k$
4. ab
5. $y - 10$
6. $\frac{p}{6}$

Set C

1. $2(x + 4)$
2. t^2
3. $3w - 7$
4. $\frac{m - 1}{2}$
5. $n - 1$
6. $\frac{a + b}{2}$

Set D

1. $3n$ dollars
2. $2s + 4$
3. $2L + 10$
4. $50 - 2.5k$

Extension

1. $3e + 6$
2. $3n + 3 = 3(n + 1)$

Exit ticket

1. Level 1: $N + 12$
2. Level 2: $y - 4$
3. Level 3: $5m + 2$

5.05 Substitution (booklet p. 19)

WE DO examples

1. Ex 1: 7
2. Ex 2: 0
3. Ex 3: 60
4. Ex 4: 109
5. Ex 5: -11
6. Ex 6: 94

Set A

1. 7
2. 4
3. 20
4. 30
5. 13
6. 0

Set B

1. 6
2. 60
3. 36
4. 3
5. 14
6. 19

Set C

1. 109
2. -11
3. 169
4. 94
5. 19
6. 319

Set D

1. 22
2. 35
3. 77
4. 3

Extension

1. \$59
2. 60 minutes

Exit ticket

1. Level 1: 20
2. Level 2: 13
3. Level 3: 10

5.07 Adding and subtracting terms (booklet p. 23)

WE DO examples

1. Ex 1: $7m$
2. Ex 2: $8m$
3. Ex 3: $2y$
4. Ex 4: $18uv + 4$
5. Ex 5: $13h^2 + 17h$
6. Ex 6: $2h^3 - 14h$

Set A

1. $7m$
2. $8m$
3. $6a$
4. 0
5. $9x$
6. $2y$

Set B

1. yes
2. no
3. yes
4. no
5. no
6. yes

Set C

1. $18uv + 4$
2. $7a + 4b$
3. $10x + 4$
4. $6pq$
5. $2m + 6$
6. 5

Set D

1. $13h^2 + 17h$
2. $2h^3 - 14h$
3. $8ab + 5a - b$
4. $6x + 5$

Extension

1. $6a^2 + 2a - 6$
2. $6m$

Exit ticket

1. Level 1: $9k$
2. Level 2: $3a + 9$
3. Level 3: $2x^2 - 3x$

5.08 Multiplying terms (booklet p. 27)**WE DO examples**

1. Ex 1: $12m^2$
2. Ex 2: a^4
3. Ex 3: $180v^2$
4. Ex 4: $15p^2q^2$
5. Ex 5: $21a^2$
6. Ex 6: $8m^3n^3$

Set A

1. $12m$
2. $10a$
3. ab
4. a^4
5. $7xy$
6. $12m^2$

Set B

1. $12pq$
2. $180v^2$
3. $2k^2$
4. $10a^2b$
5. $12x^3$
6. $8c^3$

Set C

1. $21a^2$
2. $-12ab$
3. $-3x^2$
4. $8m^2$
5. $-6x^3y$
6. $5t^2$

Set D

1. $15p^2q^2$
2. $8m^3n^3$
3. $6a^5$
4. $6xy \text{ cm}^2$

Extension

1. $9m^2$
2. $5xy$

Exit ticket

1. Level 1: $15k$
2. Level 2: $24a^2b$
3. Level 3: $6x^2y$

5.09 Dividing terms (booklet p. 31)**WE DO examples**

1. Ex 1: $5m$
2. Ex 2: 9
3. Ex 3: $7y$
4. Ex 4: $3x$
5. Ex 5: $\frac{16e}{5}$
6. Ex 6: $2xy$

Set A

1. $5m$
2. 9
3. $7y$
4. $4ab$
5. 4
6. $6p$

Set B

1. $3x$
2. $3m$
3. $2a$
4. $3t^2$
5. $\frac{x}{4}$
6. $\frac{2}{3b}$

Set C

1. $\frac{16e}{5}$
2. $-5n$
3. $3x$
4. $3ab$
5. $\frac{2a}{3}$
6. $-9y$

Set D

1. $2xy$
2. 3
3. m
4. $4y \text{ cm}$

Extension

1. $\frac{3m}{2}$
2. $2y$

Exit ticket

1. Level 1: $3p$
2. Level 2: $4ab$
3. Level 3: $\frac{-2x^2}{3}$

5.10 Guess, check and improve (booklet p. 35)

WE DO examples

1. Ex 1: yes
2. Ex 2: $m = 4$
3. Ex 3: $k = 0$
4. Ex 4: $d = 18$
5. Ex 5: $a = 9$
6. Ex 6: $x = 0.5$

Set A

1. yes
2. no
3. yes
4. yes
5. no
6. yes

Set B

1. $= 7$
2. $= 9$
3. $= 17$
4. $= 4$
5. $= 4$
6. $= 20$

Set C

1. $m = 4$
2. $k = 0$
3. $d = 18$
4. $a = 9$
5. $b = 7$
6. $x = 7$

Set D

1. $x = 0.5$
2. $n = 12$ (or -12)
3. $x = 5$ (or -6)
4. 8

Extension

1. $x \approx 3.7$
2. $x \approx 6.6$

Exit ticket

1. Level 1: yes
2. Level 2: $x = 7$
3. Level 3: $y = 20$

5.11 Solving one-step equations (booklet p.

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WE DO examples

1. Ex 1: $w = 35$
2. Ex 2: $e = 12$
3. Ex 3: $p = 3$
4. Ex 4: $x = 12$
5. Ex 5: $y = -7$
6. Ex 6: $k = -2.5$

Set A

1. 35
2. 12
3. 11
4. 20
5. 15
6. 20

Set B

1. 3
2. 12
3. 9
4. 45
5. 5
6. 30

Set C

1. $y = -7$
2. $k = -2.5$
3. $m = -15$
4. $x = -4$
5. $b = 3.5$
6. $q = 14$

Set D

1. $n + 19 = 45$, $n = 26$
2. $7n = 84$, $n = 12$
3. $n/4 = 11$, $n = 44$
4. $n - 23 = 58$, $n = 81$

Extension

1. $n = 4$
2. $a = 15$

Exit ticket

1. Level 1: $n = 22$
2. Level 2: $x = 7$
3. Level 3: $t = -24$

5.12 Solving two-step equations (booklet p.

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WE DO examples

1. Ex 1: $t = 2$
2. Ex 2: $y = 3$
3. Ex 3: $m = 68$
4. Ex 4: $x = 9$
5. Ex 5: $m = 10$
6. Ex 6: $x = -4$

Set A

1. 3
2. 2
3. 4
4. 3
5. 5
6. 5

Set B

1. 68
2. 9
3. 12
4. 20
5. 12
6. 10

Set C

1. $m = 10$
2. $k = 7$
3. $x = -4$
4. $x = -3$
5. $d = -3$
6. $x = 6$

Set D

1. $x = 12$
2. $y = 9.5$
3. $t = 4$
4. $n = 7$

Extension

1. $a = 4$
2. $x = 4$

Exit ticket

1. Level 1: $x = 6$
2. Level 2: $n = 18$
3. Level 3: $k = 9$

5.13 Equations and problem solving

(booklet p. 47)

WE DO examples

1. Ex 1: C: $17N = 1003$
2. Ex 2: $N = 59$
3. Ex 3: $5N = 180$, $N = 36$ tickets
4. Ex 4: $5N - 180 = 2100$, $N = 456$ tickets
5. Ex 5: $4s = 52$, $s = 13$ cm
6. Ex 6: $n + n + 1 + n + 2 = 72$: 23, 24, 25

Set A

1. $n + 8 = 20$
2. $5n = 45$
3. $n - 7 = 3$
4. $n/2 = 16$
5. $2n + 1 = 15$
6. $3n - 4 = 11$

Set B

1. 12
2. 9
3. 10
4. 32
5. 7
6. 5

Set C

1. \$2.50
2. 13 cm
3. \$138
4. 12
5. 13 km
6. 5 tickets

Set D

1. 23, 24, 25
2. length 16 m, width 8 m
3. 456 tickets
4. \$80

Extension

1. 7.5 weeks
2. 32 and 18

Exit ticket

1. Level 1: $n + 14 = 31$, $n = 17$
2. Level 2: $4b = 62$, $b = \$15.50$
3. Level 3: $2n + 9 = 41$, $n = 16$